Computational Physics

Assessment 3 out of 3

April 24, 2026

Total Marks: 25 **Duration:** 90 mins

Questions:

1. **Time Integration of an ODE** **(8 Marks)**
   Consider the ordinary differential equation:

$$\frac{dy}{dt} = -5y, \quad y(0) = 1$$

   (a) **Forward Euler Method** **(2 marks)**
   Using the forward Euler method, solve the equation numerically up to $t = 2$ using time step $\Delta t = 0.2$.
   Plot $y(t)$.

   (b) **RK4 Method** **(2 marks)**
   Solve the same problem using the fourth-order Runge–Kutta (RK4) method with the same time step $\Delta t = 0.2$.
   Plot $y(t)$ on the same graph.

   (c) **Implicit Euler Method** **(2 marks)**
   Solve the equation using the implicit Euler method with $\Delta t = 0.2$.
   Plot the result along with the previous methods.

   (d) **Error and Comparison** **(2 marks)**
   The exact solution is:

$$y(t) = e^{-5t}$$

   Compute the error at $t = 2$ for each method.
   Identify which method is most accurate.
   Briefly comment on the effect of time step on accuracy.

2. **1D Random Walk** **(5 Marks)**
   A particle performs a one-dimensional random walk starting from the origin $x = 0$. At each step, it moves:

   - one step to the right $(+1)$ with probability $p = 0.5$
   - one step to the left (-1) with probability 0.5

   (a) Simulate the random walk for $N = 100$ steps and plot one sample trajectory. **(2 marks)**

x^2

(b) Repeat the random walk for 200 runs. Using these runs, compute the mean squared displacement:

$$\langle x^2(N)\rangle$$

Verify that it scales linearly with N, i.e.,

$$\langle x^2(N)\rangle \propto N$$

(2 marks)

(c) What is the expected value of $\langle x(N)\rangle$? State the result with a brief justification. **(1 mark)**

3. **2D Brownian Dynamics:** **(6 Marks)**

A particle undergoing 2D Brownian motion is described by the equation:

$$\xi \dot{\mathbf{r}} = \boldsymbol{\eta}(t)$$

where $\boldsymbol{\eta}(t)$ is Gaussian white noise with:

$$\langle \eta_\alpha(t)\rangle = 0, \quad \langle \eta_\alpha(t)\eta_\beta(t')\rangle = 2B\delta_{\alpha\beta}\delta(t-t')$$

Here, α, β denotes the spatial direction and ξ is the friction coefficient.

Let your date of birth be **dd/mm/yyyy**. Define:

$$B = \frac{dd}{30}, \quad \xi = \frac{mm}{12}$$

(a) Simulate the motion and plot the mean squared displacement (MSD). **(2 marks)**

(b) Repeat the simulation multiple times (independent runs) and choose a number of runs such that the MSD shows convergence. Clearly state the number of runs used and explain how you determined this choice. **(2 marks)**

(c) Estimate the diffusion coefficient from the MSD and compare it with the theoretical value. **(2 marks)**

4. **1D Heat Equation:** **(6 Marks)**

Consider the one-dimensional heat equation:

$$\frac{\partial T}{\partial t} = D\frac{\partial^2 T}{\partial x^2}$$

where D = 0.005 m^2/s. The spatial domain is $0 \le x \le 5$ meters, with boundary conditions:

$$T(0,t) = 0°\text{C}, \quad T(5,t) = 0°\text{C}$$

The initial temperature profile is given by:

$$T(x,0) = T_0 \exp\left(-\frac{(x-2.5)^2}{0.0002}\right), \quad T_0 = 1°\text{C}$$

(a) Perform the simulation for up to a total time of 20 seconds. Plot the temperature profile $T(x,t)$ at times $t = 0, 2, 5, 10, 15$, and 20 seconds. **(4 marks)**

(b) Plot the temperature at the center of the domain, $T(x = 2.5, t)$, as a function of time. Provide a brief explanation of the observed trend in terms of heat diffusion. **(2 Marks)**